

Efficient anarchy

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Received: 23 February 2005 / Accepted: 20 June 2006
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Abstract Can anarchy be efficient? This paper argues that for reasons of efficiency, rational, wealth-maximizing agents may actually choose statelessness over government in some cases. Where markets are sufficiently thin or where government is prohibitively costly, anarchy is the efficient mode of social organization. If total social wealth under conditions of relatively lower levels of trade is not substantially smaller than it is under conditions of relatively higher levels of trade, the cost of government may exceed the social benefits it provides. Likewise, if the cost of a state is sufficiently large, even substantial differences in social wealth under these two scenarios may prove too small to justify the formation of government from a cost-benefit perspective. The framework I provide explains the persistence of anarchy in two major areas where we tend to observe it: among primitive societies and at the global level. (*JEL* P48)

Keywords Anarchy · Social efficiency · Institutions · Nuer · International trade

1. Introduction

Can anarchy be efficient? Conventional wisdom emphatically answers no. By providing centralized¹ enforcement, government enables individuals to realize gains from exchange they could not capture if the state were absent. Rationally self-interested agents therefore choose to form government. This rationale for the state is at least as old as Hobbes but remains alive and well in modern social science. Economics is no exception. As Nobel Prize winner Doug North put it: “Throughout history, individuals given a choice between a state – no matter how exploitative it might be – and anarchy, have decided for the former” (1981: 24). Political economists have accepted the efficiency of government in organizing society

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¹ Throughout this paper I use the term “centralized” synonymously with the term “state.” Centralized institutions (enforcement, rules, etc.) stand in contrast to those I call “private,” which refer to non-state institutions (enforcement, rules, etc.).

without question.² Is it possible that in some cases anarchy is actually optimal from the standpoint of social wealth?

The ubiquity of government today causes us to forget that numerous societies were stateless for most of their histories and that many remained so well into the 20th century. Some of these groups encompassed significant numbers of people. Consider, for instance, African groups such as the Tiv, which included over one million individuals, the Nuer whose population has been estimated at 400,000, or the Lugbara with over 300,000 members. More striking yet is the fact that, globally, the world has and continues to operate in the context of “international anarchy.” The continued presence of numerous sovereigns creates massive ungoverned interstices for many of the interactions between the inhabitants of these different nations.

The observed absence of government in these environments requires explanation. If Hobbes and the generations of economists who have followed him are correct, anarchy is inefficient and government should have quickly replaced vacuums of centralized authority. But why then did statelessness among numerous societies last so long? For that matter, what accounts for the continued existence of international anarchy? In short, how do we explain the persistence of significant arenas of anarchy over time?³

North (1990) suggests that inefficient forms of social organization may persist because of path dependence. Since in *most* instances where we observe the absence of effective state enforcement (for example, in many transitioning economies) this arrangement is inefficient, it is tempting to conclude that in *all* instances where we observe this arrangement that this is so. It would therefore be very easy to mistakenly dismiss all instances of anarchy’s persistence by reference to path dependence. In contrast, this paper will explore why it may actually be rational in some cases for wealth-maximizing individuals to choose *not* to form any government at all.

There are two general explanations for government’s presence – social contract theory and predatory theory. The former suggests that agents choose to form government because it is socially efficient. Individuals recognize that by introducing the state they can move from a situation of conflict (or lesser cooperation) to one of greater cooperation (see Buchanan, 1975; Buchanan & Brennan, 1980; North, 1981; North & Thomas, 1973). The predatory theory of government on the other hand suggests that the state emerges out of the self-interested behavior of some agent (or group of agents) with a comparative advantage in using force (see Olson, 1993; McGuire & Olson, 1996).⁴

This paper employs the social contract theory of government (or in this case, absence of government) described above. I do not take this theory to be an accurate description of the actual emergence of governments nor do I believe that governments consist of benevolent, social welfare-maximizing agents and do not suffer from the standard problems of public choice.⁵ Nevertheless, I adopt the contract theory of government because it offers the strongest

²David Friedman (1973), Murray Rothbard (1977), Bruce Benson (1999b), and Randy Holcombe (2004) are rare exceptions in this regard.

³While no one has addressed this question, a small but growing literature deals broadly with the economics of anarchy. See for instance, Dixit (2003, 2004) Hirshleifer (1994), Bates et al. (2002). Dixit’s (2003) analysis is probably the most closely related to mine in that it considers the limits of self-governance and when centralized governance becomes efficient.

⁴Levi (1988) points out that consent and predatory theories of the state are not mutually exclusive, since in either event a ruler requires the consent of at least some people to govern. While I accept this important insight, it is nevertheless possible to conceptually distinguish between these theories on the grounds outlined above.

⁵It is important to emphasize that assuming away predatory behavior on the part of government for my analysis is purely for the purposes of addressing the ‘hardest case’ for anarchy – when government only aims

justification for the state and thus allows me to demonstrate that, even in this case, there are important conditions under which anarchy is the socially efficient arrangement.

Section 2 of this paper considers the determinants of anarchy's efficiency. It looks at what factors impact the cost of having government and what factors impact the benefit government provides by moving society from a lower trade equilibrium to a higher trade one.⁶ Section 3 applies this framework to observed instances of anarchy. In particular it uses this framework to shed light on why we tend to see statelessness in primitive societies and on the global scale. Section 4 concludes by discussing transitions from anarchy to government.

2. When is anarchy efficient?

As previous work has shown, in the absence of government, private institutional arrangements emerge to prevent conflict and encourage cooperation (Benson, 1989; Anderson & Hill, 2004; Leeson, 2004, 2006; Greif, 1989, 1993, 2002; Ellickson, 1991, Clay, 1997; Landa, 1994; Milgrom et al., 1990; Greif et al., 1994). These arrangements, such as the use of multilateral punishment among small groups via ostracism or boycott, the emergence of conflict inhibiting social norms, and the use of arbitration organizations for international trade, operate primarily through mechanisms of reputation.

For the most part, however, reputation mechanisms successfully secure exchange without state enforcement among small, close-knit communities.⁷ Their ability to enable agents to realize the gains from trade is therefore limited (see for instance, Dixit, 2004; Greif, 1993, 2002; Zerbe & Anderson, 2001). As the traditional rationale for government suggests, by reducing the state of uncertainty that surrounds interacting with agents outside of one's social network, government can improve social wealth by enabling additional exchange.

In a simple two-person model of exchange, let H be the sum of the payoffs to each individual of trade when government is present and let L be the sum of the individual payoffs of the relatively lower level of trade when government is absent where, $H > L > 0$. Individuals may coordinate either on the high trade equilibrium by introducing government, or the low trade equilibrium. For government to be efficient and rationally self-interested agents to prefer its presence, the cost of government, G , must be smaller than the benefits it provides. The benefit of government is the difference between social wealth in the two states of the world described above – that in which government exists and agents are in the higher trade equilibrium, and that which it does not and agents are in the lower trade equilibrium. Government is therefore an efficient solution to the social dilemma if and only if $G < H - L$.

at enhancing social wealth. As a factual matter, both theory and evidence strongly suggest that political agents are self-interested and engage in predatory behaviors. In other words, although it is not considered here, the cost of government includes the potential for large "public choice costs," stemming from predatory political activities. For an excellent discussion of the public choice costs of government in its capacity as definer and enforcer of property rights see Anderson and McChesney (2003). Also, see de Soto (1989) who examines the outcomes of predatory governments on trade and social welfare. Finally, for a discussion of public choice costs of government in the context of international law see Benson (1999a).

⁶ The paper employs cost/benefit considerations to explain/predict statelessness. Libecap's (2003) excellent work uses a similar approach to explain/predict when individuals contract for property rights where the state is absent. Following up on this, Haddock's (2003) important work considers the use of force to establish such rights where contracting for them is prohibitively expensive.

⁷ Klein's (1992) work on credit ratings suggests that reputation can be effective in large groups. Also on the effectiveness of reputation in large groups see Leeson (2004). However, the applicability of reputation mechanisms among large populations is overwhelmingly rejected in the literature that discusses its application. See for instance, Dixit (2004), Greif (1993, 2002) and Zerbe and Anderson (2001) among others.

Where $G > H - L$, anarchy is efficient. Whether G is actually greater than $H - L$ obviously depends upon two things. First, the size of G , and second the size of the gap between social wealth when agents are in the higher trade equilibrium vs. when they are in the lower trade equilibrium.

2.1. The cost of government

The cost of government can be broken into three primary components: (1) first, there is a simple organizational cost of creating a state – the cost of organizing collective action. Concretely, the organizational costs of government include (a) the decision-making costs of arriving at the specific set of rules the state is to enforce and (b) the external costs of collective decisionmaking, which result from the fact that the group may sometimes make choices that are contrary to the interests of the individual (Buchanan & Tullock 1962). The organizational cost of government thus depends upon, in addition to other possible factors, the form of government or decision-making process that is followed in determining what set of rules the state is to enforce.

- (2) The second cost of government is the cost of enforcing decided upon rules. These costs are expenditures associated with creating and maintaining police and military forces, and a court system. Enforcement costs are increasing in population size, as it is more expensive to police 1000 people, for instance, than it is to police 10. Besides population size, the enforcement costs of government are also increasing in population heterogeneity. Ethnically, religiously, linguistically, and otherwise fractionalized populations are more prone to disagreement, mistrust, and violent conflict than those that are less fractionalized (see for instance, Alesina et al., 2003; Alesina & Spolaore, 2003; Alesina & La Ferrara, 2002). The state's enforcement entities – for instance the police and courts – are therefore deployed more frequently for the purposes of preventing and settling disputes among socially disparate populations than among more homogeneous ones. Finally, the form of government may influence enforcement costs as well.
- (3) The third cost of government is the cost of providing public goods other than those necessary to enforce decided upon rules (such as police and courts, which falls under the enforcement costs of government in (2)), but which contribute to the ability of individuals to engage in higher levels of trade.⁸ Roads, for instance, which permit interactions between larger numbers of individuals, are an example of this. The public goods cost of government is largely determined by the same factors as enforcement costs. Public goods costs are increasing in the size of the population because it is more expensive to supply an adequate system of transportation for a large population, for instance, than a small one. Similarly, *ceteris paribus*, public goods costs will be higher among socially heterogeneous populations than more homogeneous ones (Alesina, Baqir & Easterly, 1999). Where agents have more diverse characteristics they have more diverse needs, requiring multiple forms of the same public goods (for instance roadways coming from, and going to, different places), which raises the cost of providing such services.

⁸These goods need not be non-excludable and non-rivalrous, per the classic definition of public goods. Roads, for example, are both excludable and rivalrous, and yet government is traditionally viewed as the appropriate provider of transportation systems. An important strand of work points out that many if not most of the public goods traditionally thought of as within the purview of state provision (including roads) can actually be provided for privately and have been historically. See for instance, Bieto et al. (2002), which contains a number of examples of this. Here, however, I am taking the more traditional view held by most economists, which assumes that government will provide these goods.

2.2. The benefit of government

The factors above determine G 's size. The efficiency of anarchy, however, depends upon the cost of government relative to the benefit that government provides by moving society to the higher trade equilibrium. What then affects the size of $H - L$?

The difference between social wealth when individuals engage in higher trade vs. when they engage in lower trade is determined by the potential for gains from exchange. The size of these gains is in turn a function of the range of exchange opportunities that are available to them. Five main factors affect this range:

- (1) Individuals' endowments – *Ceteris paribus*, where agents begin with more disparate endowments the gains from trading will be larger and vice versa.
- (2) The size of the potential trading population – A larger population of potential exchange partners means a larger number of opportunities to gain from trading. A smaller population means fewer potential gains from exchange.
- (3) Individuals' productive abilities – *Ceteris paribus*, where individuals' productive abilities are more disparate there are larger gains from them exchanging. Where productive abilities are more similar the opposite is true.
- (4) Individuals' preferences – *Ceteris paribus*, more diverse agent preferences create more opportunities for exchange. Less diverse preferences mean fewer opportunities from exchange.
- (5) The presence or absence of private institutional arrangements that facilitate exchange – Where private institutions, like arbitration, reputation mechanisms, or community norms compelling cooperation are present, agents are able to realize additional gains from exchange. Thus social wealth in the lower trade equilibrium will be higher than it would have been without these institutions (though because of the limitations discussed above, still lower than if government existed). The presence of private institutional arrangements facilitating exchange thus shrinks the gap between social wealth in the higher and lower trade equilibrium. The absence of such institutions increases this gap.

These five factors together determine the thickness of the market. Thick markets have many (i.e., widespread) opportunities for exchange and thus generate high gains from trade. Thin markets, in contrast, have very few opportunities for exchange and thus generate minimal gains from trade.

It should be clear that when markets are sufficiently thin the relative difference in social wealth between a situation in which agents engage in higher trade and a situation in which they engage in lower trade is negligible. This corresponds to the case when $H - L$ is very small. Alternatively, when markets are very thick this difference will be large.

Having established what affects the cost of government and what affects the benefits government provides by moving society from a lower trade equilibrium to a higher trade one, it is now possible to distinguish two types of efficient anarchy: (1) "big G anarchy," in which despite the presence of a substantial gap between social wealth in the higher vs. lower trade equilibrium, government is too costly to justify its emergence, and (2) "small $H - L$ anarchy," in which even though government may be inexpensive to create, the difference between social wealth in the higher and lower trade equilibrium is so small as to make the state inefficient on cost-benefit grounds. At least theoretically then, these are situations in which statelessness is socially optimal. A society of rationally self-interested agents operating in either environment would thus (rationally) choose anarchy over government.

This framework therefore predicts anarchy in two distinct sets of circumstances: one in which the costliness of the state prevents government from emerging (big G anarchy) and

one in which the absence of trading opportunities makes the benefit of introducing the state prohibitively small (small $H - L$ anarchy). I explore the evidence supporting this prediction below.

3. Two archetypes of actual anarchy

3.1. Small $H - L$ anarchy

The first archetype of statelessness – small $H - L$ anarchy – is characteristic of statelessness observed in small, primitive societies.⁹ The historical presence of long-standing, primitive anarchic societies spans the entire globe. Consider, for example, societies such as the Eskimo tribes of the North American Arctic (Hoebel, 1954), Pygmies in Zaire (Turnbull, 1961), Indian tribes like the Yoruk of North America (Benson, 1989), the Ifugao of the Philippines (Barton, 1967), the Massims of East Papuo-Melanesia (Landa, 1994), Indian tribes of South America like the Kuikuru (Dole 1966), the Kabyle Berbers of Algeria, the Land Dyaks of Sarawak and the tribal Santals of India (Barclay, 1990), none of which had governments. In several cases primitive anarchic societies remained as such well into the 20th century. The Kapauku society of West New Guinea, for instance, was stateless until about 1960 (Pospisil, 1963).

In his classic anthropological work, E. E. Evans-Pritchard (1980 [1940]) described the Nuer society of the southern Sudan circa the 1930s. The Nuer people were not alone in pre-colonial Africa in rejecting government. Inside Africa, the Barabaig, Dinka, Jie, Karamojong, Turkana, Tiv, Lugbara, Konkomba, Plateau Tonga and others all long stood as stateless or near-anarchic orders as well.¹⁰ The Nuer, however, is among the best studied of these groups and in many ways typifies general features found among other primitive anarchic societies. For this reason I consider the Nuer exclusively here, though it should be kept in mind that the lessons of this analysis apply generally to other primitive anarchic societies with similar characteristics, as I will try and highlight below.

Primitive societies like the Nuer represent instances in which rationally self-interested individuals choose anarchy over government because the difference between social wealth in the higher trade and lower trade equilibrium is extremely small. Since the formation of even the leanest government involves some fixed cost and this cost is not insignificant, a very small $H - L$ is enough make anarchy the efficient pattern of social organization.

The small gap between payoffs from higher and lower trade in primitive societies is a function of five main factors, which tend to make potential markets inside of them extremely thin:

- (1) These societies are typically rather small, meaning there are relatively few opportunities for exchange even if government is introduced. This tends to make lower levels of trade enabled by private institutional arrangements not much less profitable than higher levels of trade that presumably would be made possible if government were established.¹¹ The size of the relevant trading population is largely determined by the size of the population over

⁹This is not to say, however, that anarchy is only efficient and therefore only arises in situations where H and L are themselves small. $H - L$ can also be small when H and L are both large. For instance, as discussed below, modern international has a large L because of highly effective private institutions. Thus, adding government will not add substantially to trade, making $H - L$ small despite the fact that H and L are themselves large.

¹⁰For reference to these and other stateless societies-in Africa see Bohannon (1968) and Barclay (1990).

¹¹I say presumably here because it is not at all clear that government would actually enable a higher level of trade in these societies. There are many examples of tribal societies that have had governments imposed

which government is introduced. Thus to understand why the relevant potential trading population for the Nuer was very small, we need to first understand why government among the Nuer, if it were introduced, would have been introduced at a low level – i.e., over a small population.

The Nuer were actually one the largest primitive stateless societies.¹² The most liberal estimate of the Nuer population is around 400,000 individuals (Barclay, 1990). However, Evans-Pritchard, who studied the Nuer most closely, estimated the Nuer population at only half this size. This figure is inclusive of all Nuer group members. This inclusive population was divided into 11 tribes: the Bul, Leek, western Jikany, Nyuong, Dok, Jagei, Gaawar, Thiang, Lak, Lou, and eastern Jiknay. Each tribe was in turn subdivided into numerous sections based on lineage, and these sections were further subdivided into numerous village communities. Nuer communities tended to be extremely close-knit, as they were composed of individuals connected by lineage.

The largest Nuer organizational unit in which private rules and arbitration procedures were respected by other individuals was the tribe. Beyond the bounds of each tribe there was no recognition of such rules or procedures. The largest conceivable level at which government might have been introduced among the Nuer would therefore have been the tribal level. Even this, though, is questionable. Evans-Pritchard, for instance, indicates that in many cases the largest effective organizational unit of the Nuer was actually much smaller, perhaps somewhere between the village and tribal levels. This suggests that were government introduced, it would have been over an even smaller population. In any event, the relevant population of the Nuer, which as already noted was considerably larger than many other anarchic primitive societies, was not very large. The presence of private institutions such as the “leopard-skin chief” who arbitrated disagreements among tribe members enabled some degree of exchange between individuals at this level. While introducing government at the tribal level might increase this exchange somewhat, the small population involved coupled with private institutions like the leopard-skin chief suggests that this increase would be minimal.

- (2) Individuals in primitive societies typically have very similar productive abilities. Most are either pastoral or horticultural. The Nuer, who were of the pastoral variety, were overwhelmingly a cattle-herding people. Though they were sometimes forced to raise crops (for instance, when rinderpest destroyed their livestock), a combination of the natural environment they found themselves in and Nuer culture created a situation in which there was very little differentiation in agents’ productive capacities.

Evans-Pritchard described the fundamental environmental features of Nuerland as follows: “(1) It is dead flat. (2) It has clay soils. (3) It is very thinly and sporadically wooded. (4) It is covered with high grasses in the rains. (5) It is subject to heavy rainfall. (6) It is traversed by large rivers that flood annually. (7) When the rains cease and the rivers fall it is subject to severe draught” (1980: 55). While these conditions allowed for occasional horticulture, hunting and fishing, they overwhelmingly dictated the productive activity of cattle herding that the Nuer were so fond of. Production was thus almost exclusively directed at raising cattle for meat and milk.

Nuer culture, which was from top to bottom organized around the importance of cattle, reinforced herding as the virtually exclusive productive activity of the Nuer people. As

on them by colonizers, and then following the withdrawal of the colonial government, have economically crashed. In these societies, trade was more expansive under statelessness than it was following the imposition of government. Leeson (2005) examines such cases in precolonial/postcolonial Africa.

¹²To my knowledge, only the Tiv and the Lugbara were larger.

Evans-Pritchard put it: nearly all of Nuer “social behavior directly concerns their cattle” (1980: 18). This fact was manifest in practices and institutions among the Nuer from the giving of names (which were based on the names of family cattle), their networks of kinship ties (which were based upon cattle ownership), to rituals and religious activities. This intensely focused interest on cattle that was fundamental to Nuer culture strengthened the singularly directed aims of Nuer productive activities in herding.¹³ Neither this feature of Nuer life nor the fact that the Nuer environment was not suitable for much other than cattle herding is to deny that innately occurring comparative advantage among individuals in say the production of milk vs. the production of meat allowed for some specialization. However, for the reasons discussed above, the degree of this specialization was severely limited. This in turn limited the gains that could be had from higher levels of exchange, which introducing government might bring.

- (3) Not always, but frequently, the agents who populate primitive societies have homogeneous preferences. In the case of the Nuer, as mentioned above, this preference was nearly uni-dimensional and aimed at the ownership of cattle. This lack of diversity tended to diminish the increase in exchange opportunities that introducing government might bring. As Evans-Pritchard observed, the “Nuer have nothing to trade except their cattle and have no inclination to dispose of these; all they greatly desire are more cattle . . . This narrow focus of interest causes them to be inattentive to the products of other people, for which, indeed, they feel no need and often enough show contempt” (1980: 88).¹⁴
- (4) Individuals in primitive societies often have very similar endowments. Because they are frequently egalitarian, these societies create a situation in which across current members and even generations, individuals have the same level and forms of wealth. For the Nuer, while disparities in wealth levels were permitted, the form of individual endowments was nearly identical for all individuals in that wealth was construed almost exclusively in the form of cattle, which was singularly desired.
- (5) Because of their small, close-knit nature, primitive societies are often able to effectively use private institutional arrangements based on norms and reputation to facilitate cooperation. Within the same communities, for instance, the Nuer shared common norms regarding the settlement of disputes, which typically involved cattle. Disagreeing members would see the leopard-skin chief who, sometimes in conjunction with community elders, would recommend how the dispute should be settled. This form of private arbitration enabled community interaction despite the absence of government.

Similarly, within the same tribe, the institution of the feud, which involved specific steps for dealing with more serious transgressions, was respected by the Nuer and provided a strong incentive for individuals to refrain from theft and violence.¹⁵ The strong presence of these private mechanisms of governance raised the relative payoff to indi-

¹³ The Nuer focus on cattle was both a cause and consequence of the cultural characteristics described.

¹⁴ To satisfy basic dietary requirements to sustain life, the Nuer were forced to spend some time in agricultural activity directed at producing carbohydrate-rich foods. In their case this was grain. However, as discussed above, for reasons of climate and preferences, the time they spent doing so was minimal. Evans-Pritchard described the situation as follows: “(1) that Nuer cultivate only enough grain for it to be one element in their food-supply and not enough to live on it alone; (2) that with their present climate and technology considerable increase in horticulture would be unprofitable; and (3) that the dominance of the pastoral value over horticultural interests is in accord with oecological relations which favour cattle husbandry at the expense of horticulture” (1980: 81).

¹⁵ For an excellent analysis of the feud and leopard-skin chief as institutions of self-enforcement see Bates (1983).

viduals in the lower trade equilibrium, which served to shrink the gap between social wealth in this and the higher trade equilibrium and with it the benefit of introducing a state.

3.2. Big G anarchy

Anarchy may also be efficient if the cost of government is extremely large. In this case even a substantial gap between social wealth in the higher vs. lower trade equilibrium may not be large enough to make the state efficient from a cost-benefit perspective. In this environment, rationally self-interested agents will again be led to prefer anarchy over government. Big G anarchy is thus the second archetype of statelessness I consider.

Instances of big G anarchy are less prevalent than instances of small $H - L$ anarchy simply because the cases in which G is likely to be massive are also the cases in which government is being extended over a massive population, which means that the potential increase in gains from trading are also massive. It is therefore hard for the cost of government to be larger than the difference between social wealth in the lower vs. higher trade equilibrium.

Nevertheless, one particular instance of big G anarchy is hard to miss: international anarchy. In recent decades there has been some growth of supranational organizations aimed at increasing the degree of centralized enforcement in the international sphere. Such organizations include, for instance, the World Court and the United Nations. These organizations, however, have not fundamentally affected the anarchic nature of the international sphere in that none of them give final, ultimate authority to the governing body to offer binding decisions on the parties involved. In short, they do not override national sovereignty but instead rely fundamentally upon the willingness and voluntary consent of the various sovereigns involved. If a sovereign chooses not to appear before such a court or does not abide by the court's decision, there exists no centralized authority to compel it do otherwise.

It might be assumed that the difference between social wealth in the higher and lower trade equilibrium in the international sphere would be very substantial, given the considerable population of the world. Even if this is the case, however, the organizational and enforcement costs of a global government extending over 6.5 billion people are prohibitively expensive. Consider, for instance, the substantial increase in organizational costs that would result from most voters being far removed from their public representatives (at least at the highest level). Organizational costs would also rise considerably because of the vast increase in the heterogeneity of the relevant population. If it is difficult to arrive at a decision regarding where a new police station is to be located within a community of 20,000 suburbanites, imagine the difficulty of coming to a much larger decision when over a billion people are involved from Beirut to Mexico City. Increased heterogeneity among the relevant population will lead to substantial increases in enforcement and public goods costs for similar reasons. Indeed, as the recent research of Alesina and Spolaore (2003) shows, the attendant increase in such costs associated with extending government over larger and more socially diverse populations is a primary constraint on the effective size of nations. At the size necessary to effectively govern the entire globe, any economies of scale in having a centralized state that normally exist on the national level are overwhelmed by the diseconomies of an encompassing world state.

Although it relates to the difference between $H - L$ instead of the size of G , it must be noted that the strong presence of private institutions that facilitate exchange in the international arena also contributes to the efficiency of anarchy in this sphere. Modern-day international trade is based largely on the set of private institutions that governed such exchange

when it first emerged on a significant scale in 12th century Medieval Europe. This set of private institutions is called the *lex mercatoria*, or law merchant.¹⁶

The law merchant is a complex polycentric system of customary law that arose from the desire of traders in the late 11th century to engage in cross-cultural exchange. In the absence of state enforcement, this custom-based system relied on private arbitration for resolving disputes. Between the early 12th and late 16th centuries virtually all European trade operated this way with great success.¹⁷ This system enabled large numbers of merchants to expand trade significantly and realize substantial additional gains from international exchange (Milgrom et al., 1990).

Contemporary international trade continues to make wide use of private arbitration as a means of settling disputes. Today at least 90 percent of all international trade contracts contain arbitration clauses (Volckart & Mangles, 1999; Casella, 1996). Among the most notable arbitration organizations that exist for this purpose are the International Chamber of Commerce (ICC), the London Court of International Arbitration (LCIA), the Arbitration Association of the Stockholm Chamber of Commerce, and the American Arbitration Association's International Center for Dispute Resolution (ICDR). In 2001 nearly 1,500 parties from over 115 nations across the globe utilized the services of the ICC alone (ICC Bulletin, 2001). The amounts in dispute varied from \$50,000 to more than \$1 billion with over 60 percent of all disputes involving sums of money between \$1 million and \$1 billion (ICC Bulletin, 2002). Similarly, the ICDR arbitrated a caseload in 2001 worth more than \$10 billion involving parties from 63 countries across the globe (ICDR, 2002).

These arbitration associations rely heavily upon evolved customary law that dictates how exchange disagreements are to be settled and "arbitral awards are most generally promptly and willingly executed by business people" (David, 1985: 357). Indeed, virtually "[e]very research into the practice of international arbitration shows that by far the great majority of arbitration awards is fulfilled without the need for enforcement" (Böckstiegal, 1984: 49). In a study published in 1981, for instance, a survey of international oil traders indicated that over 88 percent of all contracts entered were carried out without dispute. Of the remaining 12 percent, respondents indicated that 76 percent of disputes were arbitrated successfully by private adjudication (Trakman, 1983: 53). The world's largest international arbitration association, the ICC, estimates that 90 percent of all its arbitral decisions are complied with voluntarily (Craig et al., 2000: 404).

The presence of private institutional arrangements like private arbitration and reliance upon customary law in the international sphere enables a substantial amount of trade despite the absence of government. Consider for a moment the staggering level of international trade. In 2003, world exports of merchandise and commercial services alone exceeded \$9 trillion (WTO, 2004). Thus, although without centralized enforcement, agents may be situated in the lower trade equilibrium, this level of trade is not very low at all and quite possibly not significantly lower than it would be if an agency of centralized enforcement were introduced. In conjunction with the fact that the cost of such an agency would be extremely high, this strongly suggests that anarchy is the most efficient way of organizing the international arena. While some attempts have been made to introduce bodies of centralized enforcement on the

¹⁶ For a discussion of the law merchant, both modern and medieval see: Mattli (2001); Volckart and Mangles (1999); Casella (1996); Benson (1989).

¹⁷ As Benson notes, "In fact, the commercial revolution of the eleventh through fifteenth centuries that ultimately led to the Renaissance and industrial revolution could not have occurred without... this system" (1990: 31).

global level, it should therefore not be particularly surprising that global anarchy continues to persist.

4. Concluding remarks: From anarchy to government

The efficiency of anarchy in some primitive stateless societies and on the international level does not mean that statelessness is always or will remain efficient in these areas. If, for example, the members of primitive societies like the Nuer decided to widen their preferences, diversify their productive activities further, be more inclusive of other groups or, what is equivalent, take an interest in interacting with a wider, more diverse population, the thickness of potential markets would grow and with it so too would the gap between social wealth in the higher and lower trade equilibrium. If this gap grows large enough, the introduction of a state may become efficient and thus prove desirable.

Clearly, a significant factor contributing to this process – enlarging the number and range of individuals agents will interact with – is partially endogenous to the presence of government. The establishment of a state may make agents feel more secure in interacting with outsiders and thus increase market thickness, which increases the benefit of having a state in the first place. This does not mean, however, that introducing government in small $H - L$ anarchies would necessarily make government efficient. In addition to the other factors affecting the distance between social wealth in the higher and lower trade equilibrium that are not endogenous to government, agents would need to desire to interact with those outside their relatively small communities.¹⁸ In the case of the Nuer, for instance, it does not seem that this was so.

In other cases, however, it was clearly the case. As Greif (2002) points out, Genoese traders initially employed primarily private institutions to facilitate trade with one another. At some point, however, they desired to interact beyond these bounds and believed that centralized arrangements were necessary to achieve this. Thus they shifted from reliance upon private institutions of enforcement to state enforcement.

Acknowledgements I am grateful to Peter Boettke, Tyler Cowen, Chris Coyne and an anonymous referee for indispensable comments and suggestions. The financial assistance of the Oloffson Weaver Fellowship is also gratefully acknowledged. I conducted this research in my time as a Visiting Fellow in the Political Economy and Government program at Harvard University.

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¹⁸In addition to this, if the benefits that introducing government creates are not immediate (or at least not completely so), agents will need to be sufficiently forward looking for the state to be profitable to adopt. If agents are sufficiently impatient, or if a significant portion of the benefits from introducing government will only come near the end of (or only after the end of) current inhabitants' lives, government will remain prohibitively costly to merit its introduction. In societies where life spans are not very long, this may present a problem. Short-lived agents will find government too costly to adopt, and their resulting failure to adopt government will in turn contribute to the short life span of the next generation, which will confront the same dilemma.

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